



EERA Joint Programme Energy Research – Status and outlook

Eberhard Diegele & Hans Seifert (KIT) Coordinators

and the members of the JP ES Management Board

*Mario Conte (ENEA), Jean-Philippe Nicolai (CEA), Dörte Laing (DLR),
Atle Harby (SINTEF), Matthias Noe (KIT), Peter Hall (U.Sheffield)*

***International Workshop on Energy Storage in the Grid:
Low, Medium and Large Scale Requirements
January 9, 2014, Barcelona***



Part I

What is EERA

- Homepage: www.eera-set.eu/index.php?index=79

- is an alliance of leading organizations in the field of energy research.
- was founded by leading European research institutes.
- **EERA is major tool to support SET-plan** by concentrating activities and resources.
- **EERA is integral part of Horizon 2020**
- **EERA is approached to represent the EU in international co-operations**

How Does it Work

EERA aims to:

identify and define Joint Programmes of research to be carried out by EERA coalitions consistent with the SET-Plan and taking into account the activities of European Technology Platforms and industry groupings;

share information and strategic plans

to help identify strengths, weaknesses, overlaps and gaps, in order to determine areas for coordinated effort;

engage proactively with industry

to create and build on partnerships of mutual interest and benefit.

Joint Programme – Policy as of today

- Participation in EERA Joint Programmes in principle open to all research organisations
- Not just a membership; need to bring in significant R&D **capacity**
- **Mainly** based on own resources
- EERA aims to be flexible & non-bureaucratic

Long term strategy and work plan

- **Agreed Objectives and Milestones**
- **Agreed Description of Work**
- **Agreed Division of Tasks and Responsibilities**
 - Context supporting specialisation of Participants
 - Virtual centres working as one team on one topic

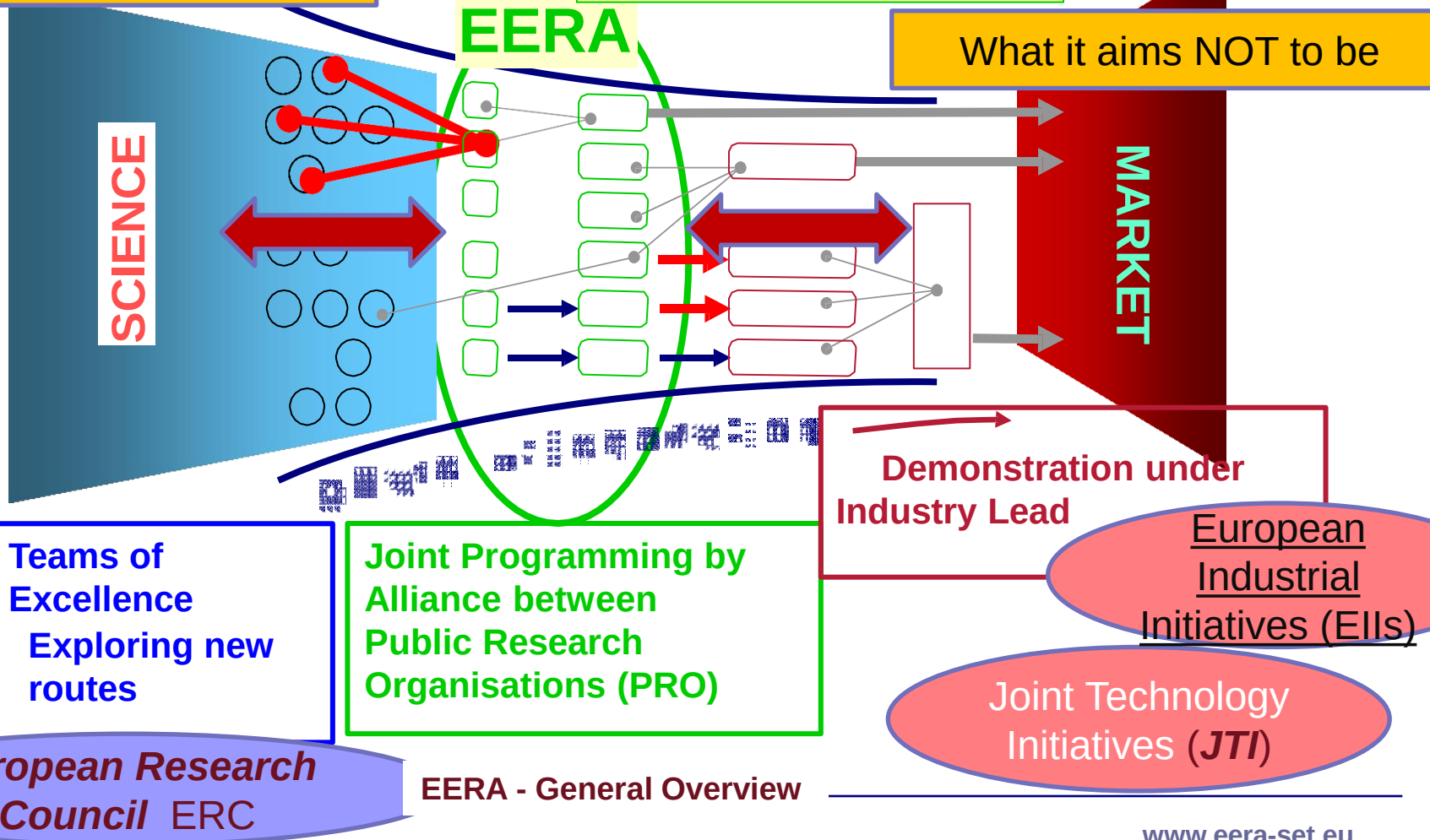
Clear and efficient coordination

Positioning EERA along the Chain of Innovation

What it aims NOT to be

Present EU system
lacks an essential
vehicle of coherence :
Alliances of PROs

What it aims NOT to be



EERA - General Overview

EERA Joint Programmes

- Joint Programmes launched 2010
 - PV: ≈ 100 professionals*
 - Wind: ≈ 130 professionals*
 - **Geothermal:** ≈ 350 professionals*
 - Smart Grids: ≈ 100 professionals*
 - Bioenergy: ≈ 120 professionals*
 - CCS: ≈ 270 professionals*
 - Mat. Nuclear: ≈ 150 professionals*

Participation increasing
New organisations joining

* Full time Equivalent



EERA Joint Programmes

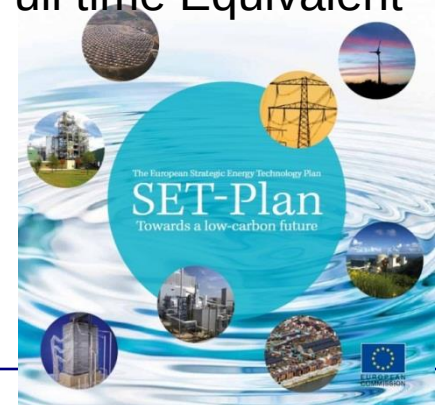
- Joint Programmes launched November 2011
 - CSP: $\approx$ 80 professionals*
 - Ocean Energy: $\approx$ 35 professionals*
 - AMPEA: $\approx$ 250 professionals*
 - FuelCell&Hydrogen $\approx$ 130 professionals*
 - **Energy Storage** $\approx$ 430 professionals*
 - Smart Cities $\approx$ 190 professionals*
- *launched April 2013*
 - *Economical, Environment and Social Impacts & Shale Gas*

Participation increasing
New organisations joining

EERA JPs it's :

- more than 3,000 FTE
- about 150 members from all around Europe
- more than 400 **Research Groups**

* Full time Equivalent





Part II

EERA Joint Programme Energy Storage

Joint Programme on Energy Storage

General technologies covered in 6 Sub-Programmes



- 1) **Electrochemical Storage** (Mario Conte, ENEA)
Lithium Ion Batteries, Super Capacitors

- 2) **Chemical Storage** (Jean-Philippe Nicolai, CEA)
Hydrogen, Methanol, Ammonia



- 3) **Thermal Storage** (Doerte Laing, DLR)
Advanced Fluids, PCM,
Thermochemical Heat Storage



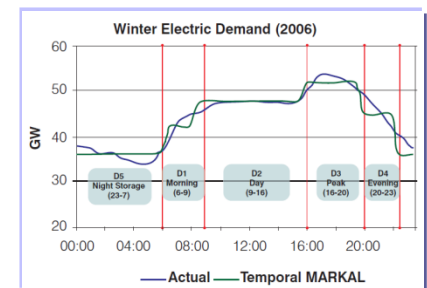
- 4) **Mechanical Storage** (Atle Harby, Sintef)
Hydro, Fly wheels, Compressed Air



- 5) **Superconducting Magnetic Energy Storage**
(Mathias Noe, KIT)



- 6) **Techno-Economics** (Peter Hall, Grant Wilson
University of Sheffield)



JP ES coordinator: Hans Jürgen Seifert, Eberhard Diegele, KIT

Partnership and Resources

Launch Nov 2011: 26 participants
from 12 EU-Member States

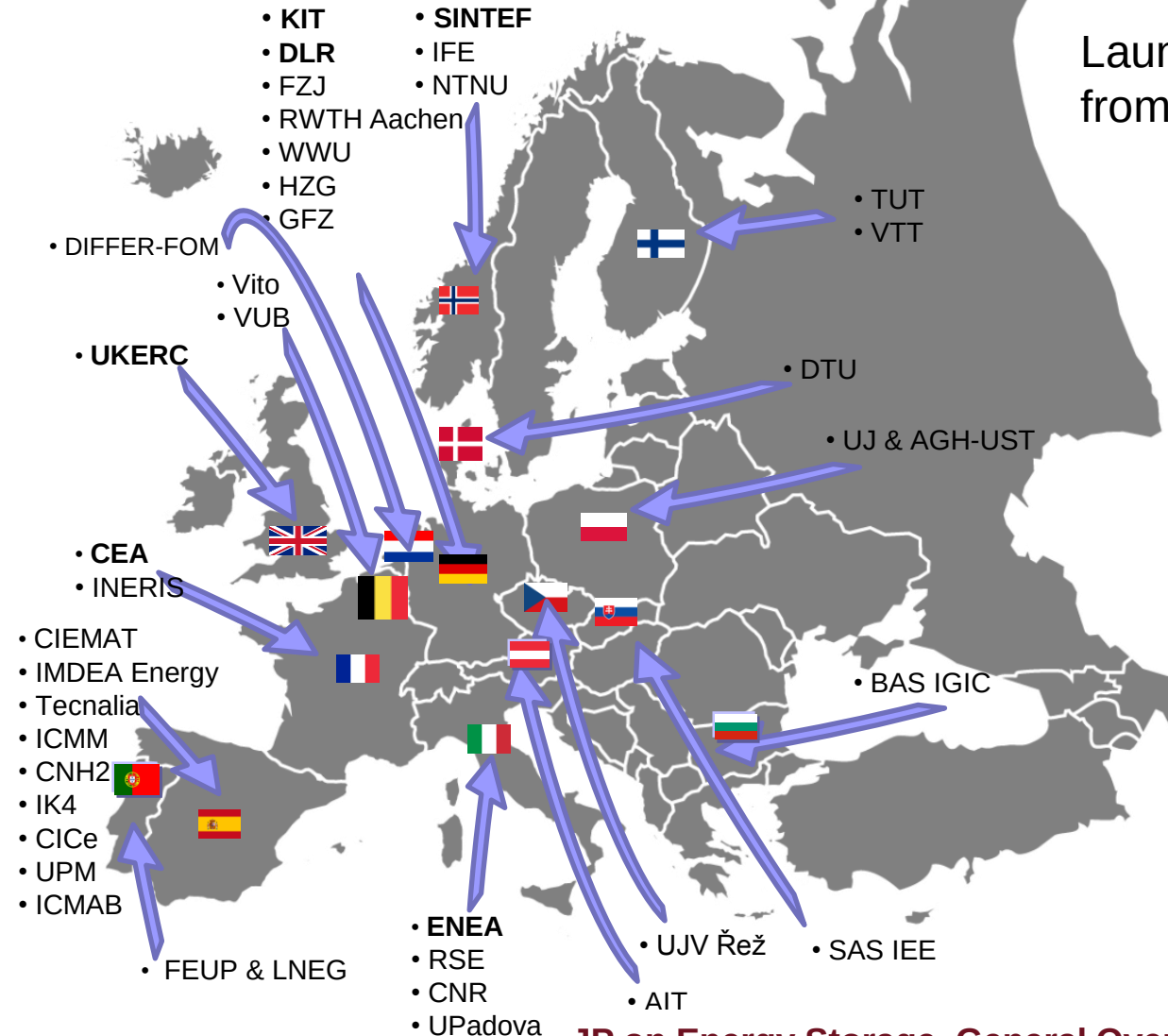
Current status 2013

- **39 participants**
- 32 full participants
- 7 associated participants
- from **16 EU member states**

Resources

committed

More than 430 PY/Y



Breakdown of Resources

Subprogramme	Electrochemical Storage	Chemical Storage	Thermal Storage	Mechanical Storage	Superconducting Magnetic	Techno-Economics
PY/Y	250	57	36	29	35	24
Participants	31	15	14	10	8	12
Countries	15	9	7	6	6	7

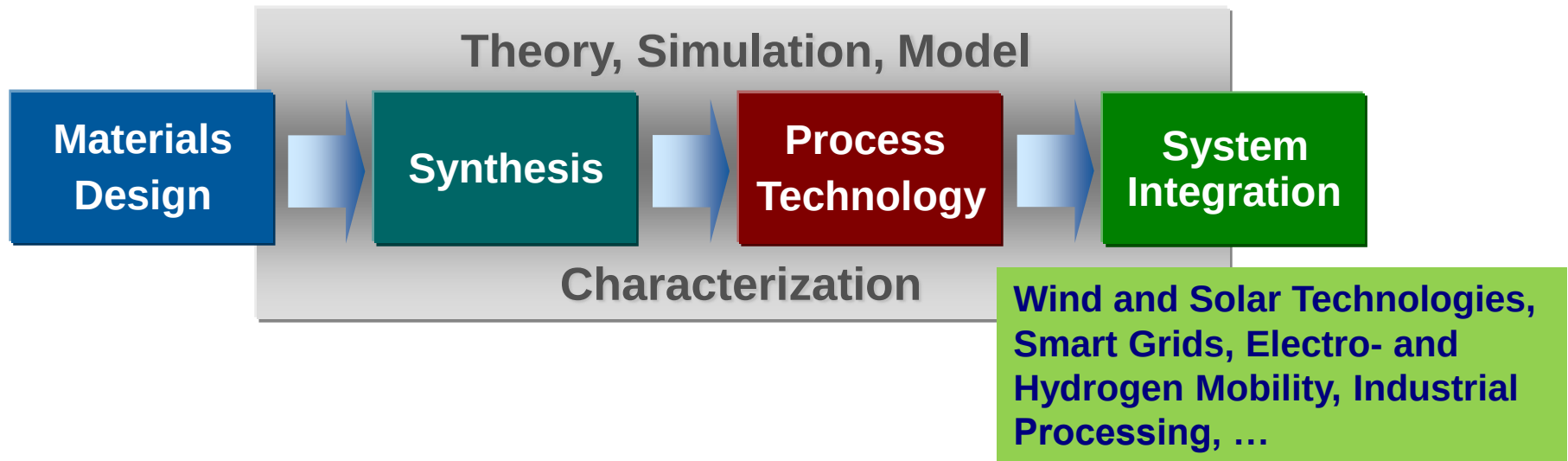
~ 60%

Total 430

The JP ES is on a good way to be established as one of the most important European research networks on energy storage

EERA JPES – Vision (one of many others)

Establishing European Platform for Integrated Energy Storage Simulation (IESS)



Comprehensive **Interdisciplinary Engineering by Design**
and use of **Modeling and Simulation** before processing
starts in the lab or in industrial operation



Achievements

Road Map as joint effort with EASE

Numerous State-of-the-art reports and summaries

Joint EASE/EERA recommendations for a
**European Energy Storage
Technology Development
Roadmap towards 2030**

*Launch event of joint
Roadmap
EERA & EASE
Brussels, 17th April 2013*



EASE: *European Association for Storage of Energy*

Electric Chain

Conventional Generation	Transmission	Distribution	Customers Services
Black start	Participation of the primary frequency control	Capacity support	End-user peak shaving
Arbitrage	Participation to the secondary frequency control	Dynamic, local voltage control	Time-of-use energy cost management
Support to conventional generation	Participation to the tertiary frequency control	Contingency Grid Support	Particular requirements in power quality
Renewable Generation	Improvement of the frequency stability of weak grids	Intentional islanding	Continuity of energy supply
DG Flexibility	Investment deferral	Reactive power compensation	Limitation of upstream disturbances
Capacity firming	Participation to angular stability	Distribution power quality	Compensation of the reactive power
Limitation of upstream perturbations		Limitation of upstream perturbations	

As identified in the Roadmap

Actual **State-of-the-Art** Repartition

Electric Chain

Technologies aggregate in focus	Conventional generation	Renewable Generation	Transmission	Distribution	Customers services
Hydro energy storage	Suitable	Possible	Suitable	Unsuitable	Unsuitable
Compressed air energy storage	Suitable	Possible	Suitable	Possible	Unsuitable
Electrochemical	Possible	Possible	Possible	Suitable	Suitable
Chemical	Possible	Possible	Possible	Unsuitable	Possible
Electrical-magnetic or Flywheels	Unsuitable	Possible	Suitable	Suitable	Unsuitable
Thermal energy storage	Unsuitable	Possible	Unsuitable	Possible	Suitable

As from Roadmap – *(Some ratings, i.e. colours, are a matter of personal view)*



Achieved Deliverables (1st year)

State of the art Reports



EERA
JOINT PROGRAMME DELIVERABLE
DRAFT REPORT

EERA Joint Programme on Energy Storage

Sub-Programme 6: Techno-Economics

M6.1 Baseline Data on existing stores of energy

EERA
JOINT PROGRAMME DELIVERABLE

EERA Joint Programme on Energy Storage
SP2 – Chemical Energy Storage

State of the Art report on gaseous and liquid hydrogen storage



EERA
JOINT PROGRAMME DELIVERABLE
DRAFT REPORT

EERA Joint Programme on Energy Storage
Sub-Programme 3: Thermal Energy Storage
M3.8 Review of currently used small scale TES systems

SP2 – Chemical Storage: State of the Art report on chemical fuels

SP2 – Chemical Storage: State of the Art report on gaseous and liquid hydrogen storage

SP3 – Thermal Storage: Review of currently used small scale TES systems

SP4 - Mechanical Storage:

Overview of the existing grid and possible future development of connections between countries in Europe

SP4 – Mechanical Storage: State of the art: Flywheels Report

SP4 – Mechanical Storage: State of the art: CAES Report

SP5 - SMES: Progress Report on “Low AC loss conductors”

SP5 - SMES: “LIQHYSMES - Hybrid Energy Storage Combining Liquefied Hydrogen and Superconducting Magnetic Energy Storage for the Large Scale Integration of Variable Renewable Energy Sources”

SP6 - Techno-Economics: Baseline Data on existing stores of energy

EERA Joint Programme on Energy Storage
Milestone 5.3 Progress Report on
“Low AC loss conductors”

“LIQHYSMES - Hybrid Energy Storage
Combining Liquefied Hydrogen and
Superconducting Magnetic Energy Storage
for the Large Scale Integration of
Variable Renewable Energy Sources”

Progress Report R5.2

Overview of the existing grid and possible
future development of connections between
countries in Europe

Deliverable M 1

EERA Joint Programme on Energy
Storage
SP2 – Chemical Storage
State of the Art report on chemical fuels

Milestone M2- Deliverable D2.1



Information on selected Sub-Programmes

- Homepage: www.eera-set.eu/index.php?index=79



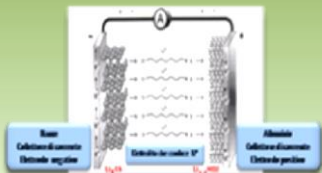
Information on selected **SP1 - Electrochemical Storage**

SP1, Electrochemical Storage - Objectives

- 1 • Creation of a strong European research network.
- 2 • Identification of key performances and specific research needs.
- 3 • Selection and investigation in a coordinated manner of key materials.
- 4 • Application of modelling and simulation.
- 5 • Formation of young researchers on material research, cell preparation and testing.
- 6 • Development of pre-normative joint horizontal activities.
- 7 • **Preparation of joint roadmaps for further research and development of materials.**
e.g. EU Integrated Energy Roadmap
Under progress this year
- 8 • **Preparation of joint research projects and selected consortia.**

Near future, after EC launch of WP 2014-15

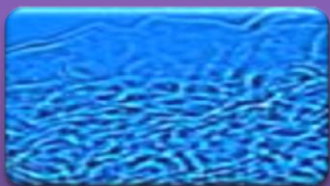
SP1, Electrochemical Storage - Work Breakdown



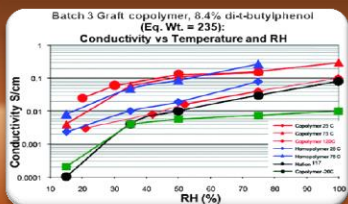
WP1: **Lithium-based batteries:** novel materials, design, integration and safety through lab characterizations



WP2: *Supercapacitors: materials, advanced design, hybrid configurations, cost reduction, characterization*

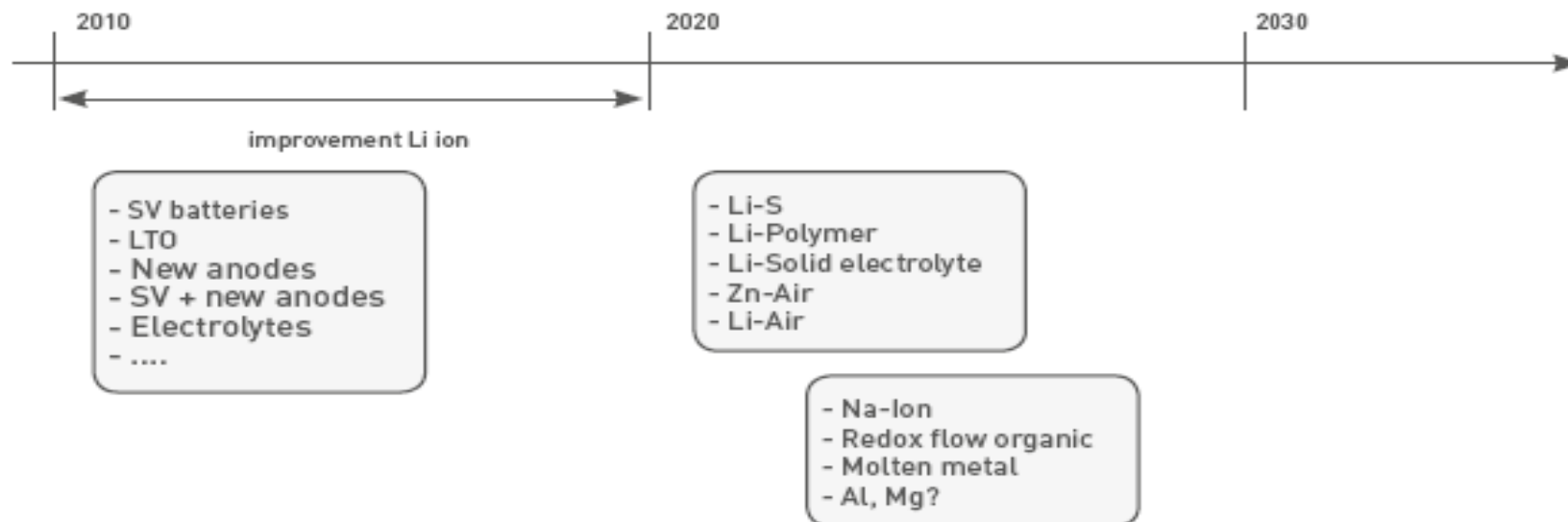


WP 3: Advanced and alternative power systems: basic materials and design study for Metal-air, Liquid Metal, Redox flow, Mg, Solid state batteries



WP4: Supporting activities: testing procedures, integrations, modelling

Batteries R&D status and future roadmap



New material needs by 2020	
LiMnNi[Co] oxides LiNiPO ₄ LiCoPO ₄ LiVPO ₄ + others	class cathode materials 4,5 - 5V
LTO C/Metal composites Si, Sn-intermetallics	class high capacity new anodes
additives electrolyte SV electrolytes	stable electrolytes

New material needs by 2030	
Li and Sulphur, improved conductive polysulphides	- Na-Ion materials - new redox flow systems materials - molten metal/salt systems - introduction of Al or Mg in metal-air, metal-sulphide or ion based systems
Development of new solid electrolytes (non-polymer)	
Development of correctly rechargeable Zn or Li-anodes, electrolytes and air cathodes	

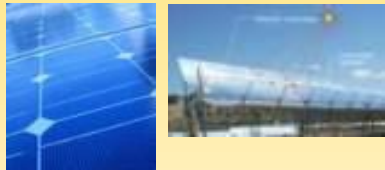


Information on selected
SP2 - Chemical Storage
Hydrogen, Methanol, Ammonia

SP2 - Chemical storage : Vision

Production

Distributed Solar
Energy Generation



Centralized Nuclear
Power Generation



Storage

Electrochemical Thermal, mechanical,
superconducting ...

Chemical storage

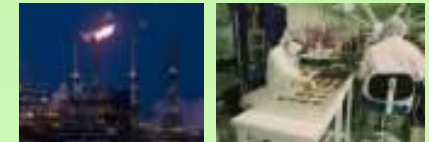
Hydrogen
(gaseous,
liquid and
solid form)



Others chemicals (NH₃,
carbon based,...)



Demand



Industry
Energy suppliers

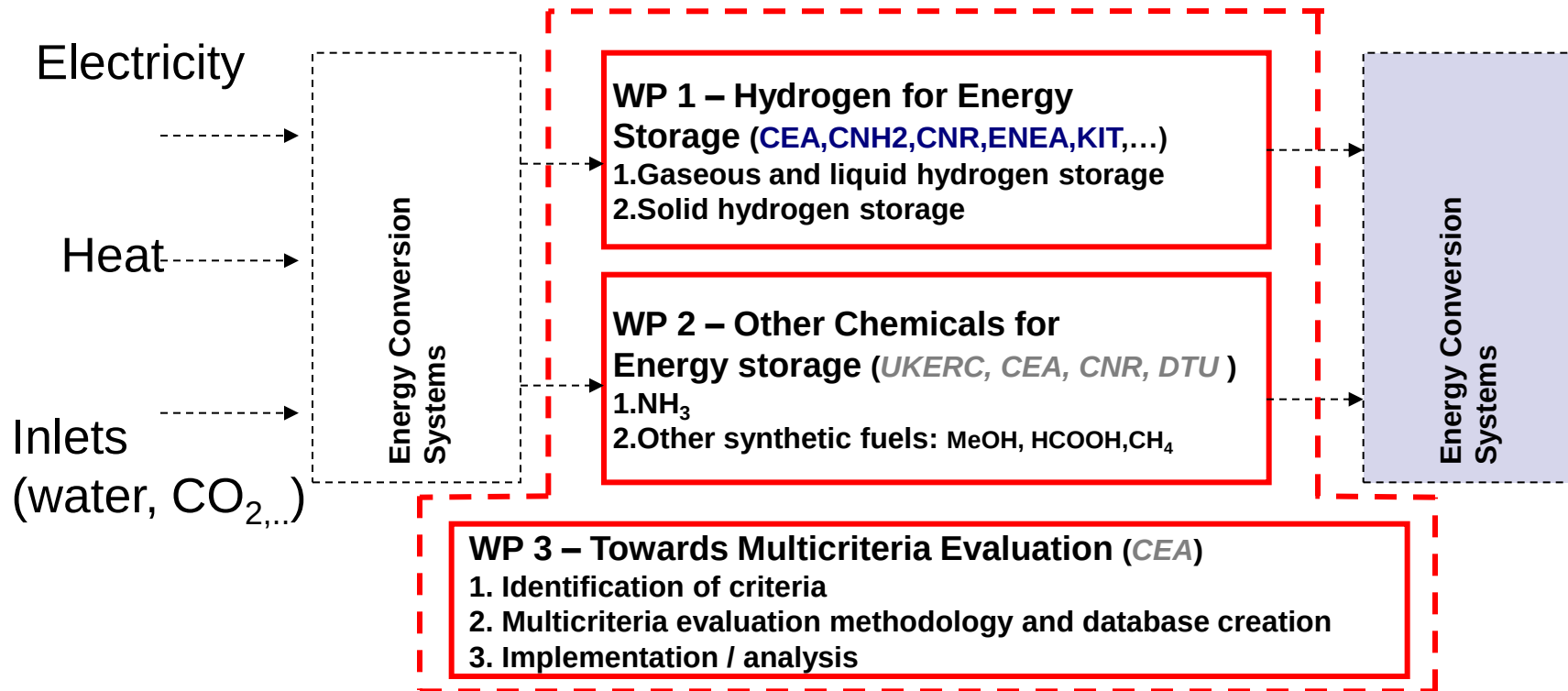


Hydrogen
& hybrid vehicles



Fuel Cells Electric, hydrogen
& hybrid vehicles

SP2 Chemical Storage – Work Breakdown





Information on selected **SP3 - Thermal Storage**

- Homepage: www.eera-set.eu/index.php?index=79

THERMAL ENERGY STORAGE

MOTIVATION

- Thermal energy storage is essential for effective and efficient generation and utilisation of heat where heat supply and heat demand do not match
- => **Key element for fuel saving and climate protection**
- Few storage technologies in high temperature range $> 100\text{ }^{\circ}\text{C}$ commercially available
- => **too expansive for broad application**
- Different applications require specific solutions concerning heat rate, capacity and system integration
- Development of innovative high temperature storage technologies required

Thermal Energy Storage

Existing high temperature storage technologies

Commercially available $> 100\text{ }^{\circ}\text{C}$:

- Hydraulic accumulator up to $150\text{ }^{\circ}\text{C}$
- Ruth's – steam accumulator ca. $100\text{--}300\text{ }^{\circ}\text{C}$
- Regenerator/Cowper Storage $> 500\text{ }^{\circ}\text{C}$
- Molten salt storage up to $400/565\text{ }^{\circ}\text{C}$



District heating,
Theiß, Austria



Steam accumulator,
Aerated concrete
manufacturing



Cowper storage,
blast furnace
industry



Molten salt storage,
Andasol power plant,
Spain

Thermal Energy Storage

Objectives

Overall objectives are:

- to provide cost effective, efficient and reliable heat storage systems
- to contribute to more wide-spread use and market penetration of the technology in the domestic, industrial and power generation sector

⇒ **cover all types of heat storage technologies:
sensible, latent and thermochemical storage**

⇒ **Main goal is cost reduction**

THERMAL ENERGY STORAGE RESEARCH NEEDS

Research Needs

- Development of advanced storage materials - sensible, latent and thermochemical - with increased energy density of storage
- Identify advanced heat transfer mechanisms for charging and discharging
- Development of advanced storage for solarthermal power plants
- Development of multi-functional materials combining heat storage, heat transfer and heat transformation
- Integration of phase change materials in building element materials
- Development of electrically heated thermal storage for PV self consumption



Information on selected **SP4 - Mechanical Storage**

***Mechanical energy storage with focus on hydropower
– status and future development***

By Atle Harby - this workshop (Thursday morning)

Compressed Air Energy Storage (CAES)

By Jihong Wang - this workshop (Friday morning)



Information on selected **SP6 - Techno-Economics**

*Evaluating the Economics, Social Benefits and Technology Needs of Energy
Storage*

By Peter Hall, Sheffield University, Sheffield. UK.

This Workshop

JP on Energy Storage, Detailed SP

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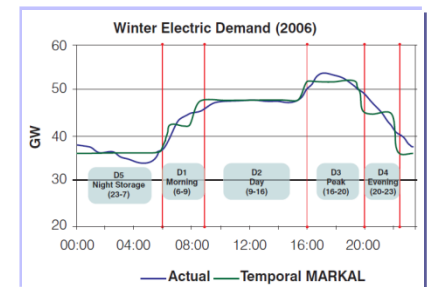
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